Dyes in History and Archaeology 21

Editor's note

Dyes in History and Archaeology is a refereed journal; the papers included are largely drawn from those presented at the annual meetings of *Dyes in History and Archaeology* and are accepted for publication, after peer review, on the assumption that a paper submitted does not contain essentially the same material as in a previous publication, and is not under consideration for publication elsewhere.

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Dyes in History and Archaeology 21

**Including papers presented at the 21st Meeting,
held at Avignon and Lauris, France, 10–12 October 2002**

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Editorial

The 21st Meeting of *Dyes in History and Archaeology* was held in Avignon and at Couleur Garance, Lauris, 10–12 October 2002, and was organised by Dominique Cardon, CNRS CIHAM-UMR 5648, Lyon, Claude Andary, Laboratoire de Botanique et Phytochimie, Faculté de Pharmacie, Université de Montpellier I, and Michel Garcia, Association Couleur Garance, Lauris. Avignon, seat of the papacy in the 14th century, is rich in historical monuments and works of art, including the Pont d'Avignon (Pont St-Bénézet) of the well-known song: a magnificent and atmospheric centre for a conference. The delegates were wonderfully entertained and treated to an extremely full and varied conference programme, amply demonstrated by the contributions published in the present volume; these represent about half of the total number of papers and posters presented. They also demonstrate the truly international nature of the field of natural dyes. Perhaps the highlight of the Meeting was the visit to Association Couleur Garance, Le Château, Lauris, visiting the old Moulin des Taillades, once used to grind madder roots, on the way; Lauris is in the Vaucluse department, formerly the heart of the French madder-growing region. Here the delegates walked in the garden of dye plants, in glorious weather (Plate 1), and watched demonstrations of dyeing. This was a splendid day, crowning a truly memorable conference and the organisers are thanked most warmly for giving the participants a most entertaining time.

I must apologise for the delay in the publication of this volume. Its preparation has been considerably disrupted by other writing and editorial commitments, including assistance on the production of the English edition of Dominique Cardon's authoritative work on natural dyes, *Natural Dyes: Sources, Tradition, Technology and Science*, published by Archetype Publications Ltd, London, in 2007. Preparation of the next volume in the series is already under way, however, and further delays are not anticipated. My thanks are due to my co-editors, Chris Cooksey, Dominique Cardon, Maarten van Bommel and Anita Quye, whose advice and support during the production of this volume have, as always, been invaluable, and particularly to Robert Atkinson, who did much of the work inserting corrections, redrawing molecular diagrams and generally helping in the preparation of the final texts. I am particularly grateful to Jim Black and Lyn Corson of Archetype Books, for their patience and cheerful encouragement.

Note: all plant names have been checked using The International Plant Names Index (2004). Published on the Internet http://www.ipni.org [accessed 3 January 2008] and other sources accessed by way of the Royal Botanic Gardens, Kew, website. Similar checks have been made on the names of animal species, fungi and lichens where these occur.

KW: Can we leave this note out? Or cut the text to save three or four lines? * The 3rd edition of Billmeyer and Saltzman's *Principles of Colour Technology*, by Roy S. Berns, published by Wiley-Interscience, appeared in 2000.

Max Saltzman: A Recollection

Joe Kirby

At the beginning of the Meeting, some words were said by Ran Boytner, University of California, Los Angeles, Dominique Cardon, CNRS, and Jo Kirby, The National Gallery, London, in memory of the eminent colour scientist Max Saltzman (1917–2001), who had died in February the previous year. One of these recollections is given below.

In 1991, the 10th Meeting of Dyes in History and Archaeology was held at the National Gallery in London. My colleague Raymond White and I sent out the invitations and then sat back to wait.

One morning, there was a telephone call. When I answered it, a voice said, 'Hullo, this is Max Saltzman. I've received your invitation and I'd like to come ...' The National Gallery Scientific Department works on many aspects of pigments and paint, including colour change. One of the best-known text books – indeed, a bible – in the field of colour science is *Principles of Colour Technology*, by Fred Billmeyer and Max Saltzman, thus when that telephone rang, we were suitably awe-struck.

Max came to London and sat for the whole afternoon having tea in our departmental library – 'No', he said, 'I don't want to see your department, thank you: I'm sure it all works fine' – and we talked for a very long time. We were extremely impressed by how well he knew our work: most of Raymond's publications have been in the field of natural products other than dyes, particularly resins, but Max knew them all.

After that DHA meeting, he managed to attend several others in spite of declining health, and he and I had many long telephone conversations after that first one.

Max Saltzman was a scientist whose opinions we valued highly. Even though he had retired, he kept his finger on the pulse, so to speak, to be aware of people, publications and research in the field of colour. His comments on experimental procedure, for example, and on the importance of the correct standards for the experiment to be performed or the sample to be investigated – textile, dyestuff or whatever it might be – were very much to the point. Apart from the book I have already mentioned,* I cannot begin to list the great number of his other, very readable, publications in the fields of dyestuffs and paint technology. And they were eminently readable: that was important to him. Max could be gruff, grumpy and demanding, like an old grandfather; he spoke his mind, and he responded to charm. For myself, I will always be grateful for his kindness and generosity with his knowledge. I treasure his memory and I am lucky to have known him. Our debt to him is incalculable.

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Spectroscopic Studies (FTIR, SIMS, ES–MS) on the Structure of Anthraquinone–Aluminium Complexes

Jana Sanyova

Abstract The structures of the complexes of alizarin and other anthraquinones (purpurin and quinizarin) with aluminium, responsible for the properties of madder lakes, have been studied using different spectrometric methods: Fourier transform infrared spectroscopy (FTIR), secondary ion mass spectrometry (SIMS) and electrospray mass spectrometry (ES–MS). FTIR unfortunately gives very little information on the structure of the complexes. SIMS and ES–MS revealed new evidence in favour of the O-1/O-9 complexation site proposed in the Kiel and Heertjes model, but suggest a different alizarin:Al stoichiometry (2:1 instead of 4:2) with possible further aggregation of these 2:1 complexes through various mechanisms.

Introduction

Alizarin (II, Fig. 1) and aluminium form the complexes which are the foundation of the colour and other properties of madder lakes. Their exact structure is still not ascertained, and, even though the model of Kiel and Heertjes with its bidentate O-1/O-9 (Fig. 2a) complexation

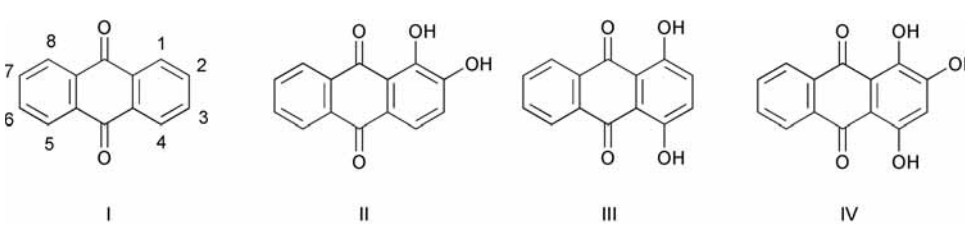

Figure 1 The structures of anthraquinone (I), alizarin (II), quinizarin (III) and purpurin (IV).

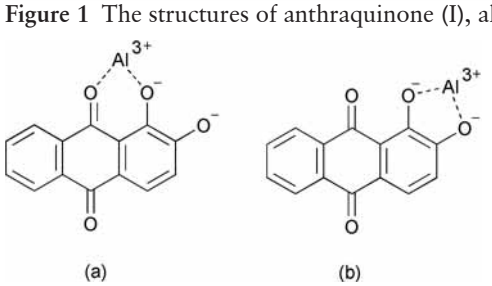

Figure 2 Two possible coordination sites: (a) O-1/O-9, as in the Kiel and Heertjes model; (b) O-1/O-2, as in the Wunderlich model.

sites is often considered as the standard structure, experimental evidence from X-ray diffraction (XRD) patterns in favour of a different coordination site (O-1/O-2) (Fig. 2b) has more recently been published.[1]

One of the reasons for this continuing uncertainty is that, with many techniques, the observable complexes are necessarily different from those in real (solid) lakes: for example, crystalline for X-rays, ionised into the gas phase for mass spectrometry. In the work presented here, anthraquinone–aluminium complexes were studied using both 'solid state' (infrared spectroscopy) and 'extractive' (mass spectrometry) methods.

Materials and methods

Lakes were prepared from synthetic alizarin (Sigma, 98%), purpurin (Sigma, 98%) and quinizarin (Extrasynthèse, 97%) using methods adapted from 17th–18th-century madder lake recipes.[2]

FTIR spectra of anthraquinones and of their aluminium lakes in potassium bromide (KBr) discs were measured using a Nicolet 5DXC spectrometer. Mass spectra of complexes observed at the surface of solid lakes were obtained using a ToF-SIMS (time-of-flight secondary ion mass spectrometry) model instrument from Charles Evans & Associates. SIMS analyses the secondary ions ejected from a small spot at the surface of a solid by a high-energy primary ion beam. The experimental conditions are such as described by Poleunis *et al.* (1997).[3]

Mass spectra of complexes formed in alizarin–aluminium solutions which had gelled were obtained using an ES–MS 'LCQ-Deca' instrument from Finnigan (Thermo-Electron Corporation). ES–MS is a milder technique than SIMS and therefore gives a better chance of observing intact complexes rather than rearranged ions. The solution of 1:2:5 aluminium: alizarin:sodium hydroxide (NaOH), using 50 mM alizarin, which had turned into a loose gel some days after the preparation, was analysed twice: first in the gel-forming stage, secondly after dilution 1:1 with methanol and sonication. The sample was injected at 3 μL min^{-1} (nano-electrospray), and positive- and negative-ion mode spectra were recorded in the 200–4000 amu range.[4]

Results and discussion

Fourier transform infrared spectroscopy

FTIR has the advantage of allowing the study of complexes '*in situ*' (in the solid state), but unfortunately the differences observed between the spectra of pure anthraquinone (Fig. 3a), alizarin (Fig. 3b), quinizarin (Fig. 3c) and lakes of alizarin (Fig. 3d) are inconclusive. The IR spectrum of unsubstituted anthraquinone (I, Fig. 1) shows only one carbonyl absorption band near 1673 cm^{-1} (Fig. 3a): as expected, the two carbonyls are equivalent to each other and absorb at the same wavelength.

In the spectrum of alizarin, 1,2-dihydroxyanthraquinone (II, Fig. 1), two carbonyl bands are observed, near 1660 and 1634 cm^{-1} respectively (Fig. 3b): the two carbonyls now absorb at different wavelengths, due to the difference in their positions relative to the two phenols on C-1 and C-2. Some authors have attributed the larger of these two shifts (about –39 cm^{-1}) to a hydrogen bond between the carbonyl on C-9

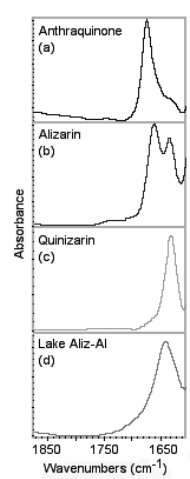

Figure 3 Infrared spectra of (a) anthraquinone; (b) alizarin; (c) quinizarin; (d) alizarin–aluminium lake, in KBr discs.

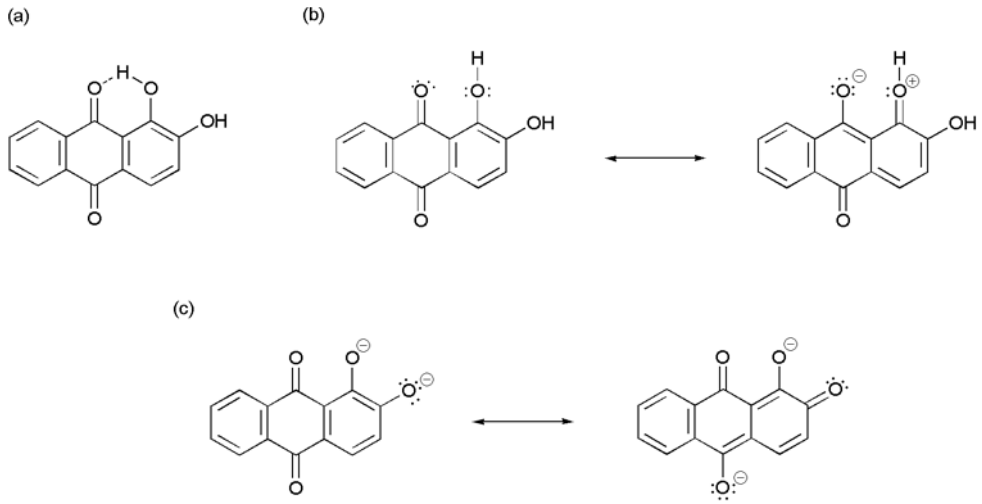

Figure 4 Possible causes for the IR band shifts: (a) 'sharing' of the proton on O-1 by the O-9 carbonyl; (b) mesomeric +M (electron donor) influence of deprotonated O-1 hydroxyl on O-9 carbonyl; (c) in aluminium–anthraquinone lakes, mesomeric +M (electron donor) influence of deprotonated O-2 hydroxyl on O-10 carbonyl.

and the phenol on C-1, but Bloom *et al*. (1959) showed that the main contribution must be the mesomeric +M electron-donor effect of the C-1 phenol on the C-9 carbonyl (Fig. 4b).[5] Their argument is based on their observation that, in the spectrum of 2-amino-anthraquinone, a large shift (from 1975 to 1625 cm^{-1}) is also observed, even though the amine on C-2 is too far removed from any of the carbonyls for hydrogen-bonding. This shift must be due to the electron donor effect of the amino group, transmitted to the C-10 carbonyl through resonance. Amines are much stronger electron donors than hydroxyls, which explains why the shifts we observed with alizarin are smaller: about -39 for the effect of the C-1 phenol on the C-9 carbonyl, and -13 cm^{-1} for that of the C-2 phenol on the further removed C-10 carbonyl.

With quinizarin, 1,4-dihydroxyanthraquinone (III, Fig. 1) and purpurin, 1,2,4-trihydroxyanthraquinone (IV, Fig. 1), only one band is observed, at about 1621 cm^{-1} (Fig. 3c). Both carbonyls, on C-9 and C-10, have now shifted to lower wavelengths, because they are in resonance with the phenols on C-1 and C-2 respectively.

The wavelengths we observed for anthraquinone, alizarin and purpurin are within 2 cm^{-1} of those reported by Bloom.[6]

In the spectrum of alizarin lakes (Fig. 3d), only one band is observed between about 1634 cm^{-1} and about 1641 cm^{-1}. The discussion above shows that this second shift, relative to alizarin, must involve the second carbonyl on C-9 and must be related to the second phenol on C-2. This is best explained by the dissociation of the second phenol into phenolate. However, this does not permit a decision to be made between the Kiel and Heertjes or the Wunderlich models for complex formation, since the phenol on C-2 becomes a phenolate in both models.[7]

Secondary ion mass spectrometry

In SIMS, the sample surface is struck by high-energy ions, and the mass/charge ratio (m/z) of the secondary ions ejected is analysed. With alizarin lakes, the main ions detected in positive and negative modes have m/z of +505 and –503 respectively (Fig. 5a, b), indicating a 2:1 alizarin:Al ratio. The –503 ion is compatible with both models (Fig. 6b, c), whereas the +505

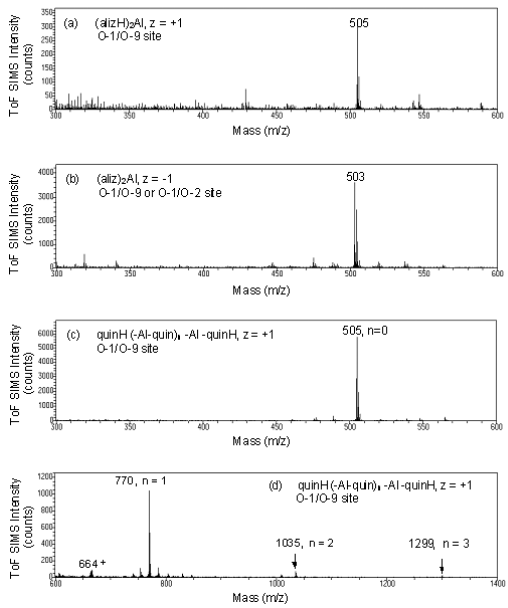

Figure 5 ToF-SIMS spectra of alizarin lakes in (a) positive and (b) negative ion mode, and of quinizarin lake in positive ion mode at (c) low and (d) medium mass ranges.

ion seems to imply that O-2 is still protonated and therefore the site for complex formation is O-1/O-9 (Fig. 6a); however both ions could also result from rearrangements under the harsh ionisation conditions. Quinizarin lakes (Fig. 6d) give a regular series of higher *m/z* ions (505 + n.265, Fig. 5 c, d), attributed to 'oligomers' qu–(Al–qu)$_n$, made possible by the presence of two equivalent sites per molecule (O-1/O-9 and O-4/O-10). The net positive charge should normally increase with the degree of polymerisation, but unexpected charges are common with SIMS.

No significant peak was obtained from purpurin lakes. It is possible that the third phenol group binds to the alumina substrate, or allows the formation of a number of non-linear complexes which are more difficult to desorb from the alumina surface.

Electrospray mass spectrometry

The milder technique ES–MS is not suitable for solids, but has the advantage that it is more

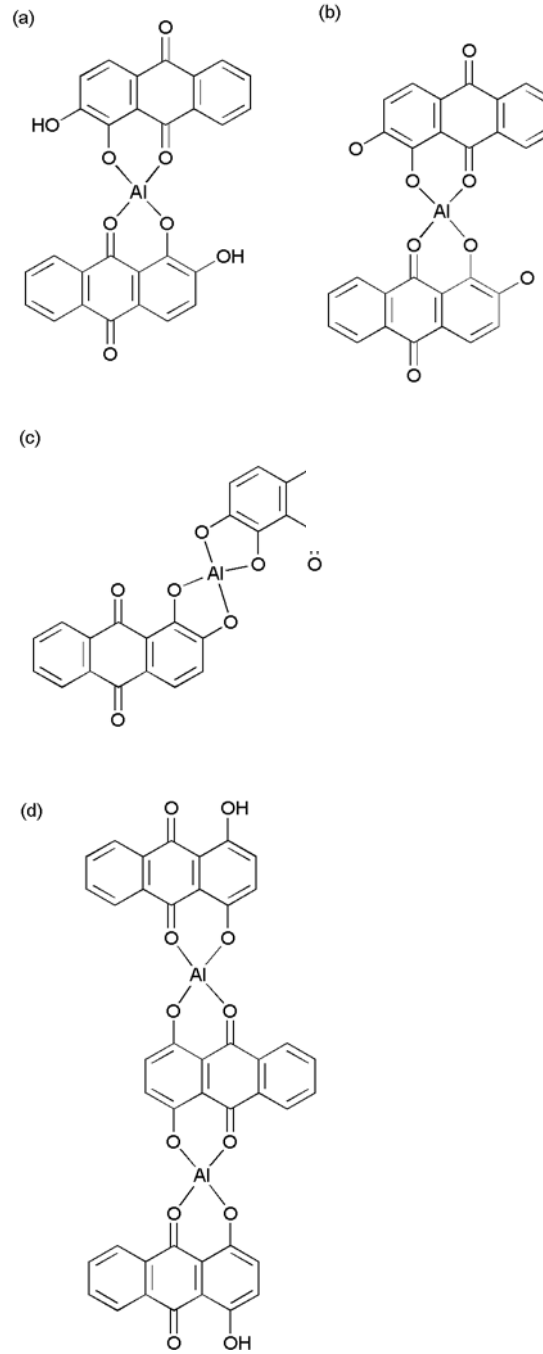

Figure 6 Possible structures for the ions detected by SIMS. (a) (alizH)$_2$Al, *m/z* = 505, z = +1, O-1/O-9 site; (b) (aliz)$_2$Al, *m/z* = 503, z = -1, O-1/O-9 site; (c) (aliz)$_2$Al, *m/z* = 503, z = -1, O-1/O-2 site; (d) (quinH-Al-quin-Al-quinH), *m/z* = 770, z = +1, O-1/O-9 site.

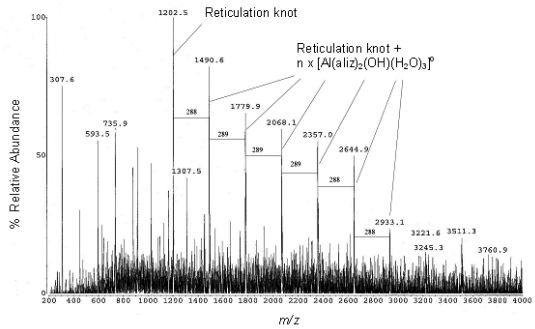

Figure 7 Positive ion mode ES–MS spectrum of a gelling alizarin–aluminium solution.

probable that the ions detected are such as they were in solution (Fig. 6). We applied it to (1:2) stoichiometric solutions of aluminium and alizarin, which were slowly turning into gels. After dilution and sonication, these solutions gave the same +505 and –503 ions as in SIMS. But unsonicated (gelling) solutions gave, in positive ion mode, a series of peaks, m/z = 1202.5 + n.288 (Fig. 7), interpreted as fragments of a three-dimensional gel network, with z = +2 and m = 2505 + n.576, where 576 is the mass of a 'monomer'. This monomer is composed of a $(alizH^{-})_2Al^{3+}$ complex (m=505) and a hydroxyl – OH^{-} – group acting as a crosslink with the next complex, plus three water molecules (Plate 51). The first peak at m/z = 2505/2 would then be the basic 'reticulation knot' of the three-dimensional network. The structure that best fits the results involves two parallel oligomeric strands, each composed of two $(alizH^{-})_2Al^{3+}$ complexes, which bond to each other by interpenetration of their aromatic side-rings.

Conclusion

Although the results presented here do not allow any definitive conclusion to be drawn about the coordination site, they are rather in favour of the O-1/O-9 site. They further suggest that, in contrast to the 4:2 stoichiometry proposed by Kiel and Heertjes, the basic stoichiometry of the complexes in the solid lakes and in concentrated solutions is 2:1, even though, in stoichiometric solutions, further aggregation of the complexes through Al–O–Al bonds and cross-linking through aromatic ring interactions are possible.

Acknowledgements

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Editor's note

This paper was presented as a poster at the Meeting. Address for correspondence: Institut Royal du Patrimoine Artistique, Parc du Cinquantenaire 1, B-1000 Brussels, Belgium. Paper received 17 March 2003; revised version received 23 February 2004.

Notes and references

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